

Jenkins

Introduction to Jenkins

Jenkins is an open-source automation server widely used in the field of Continuous Integration (CI) and Continuous Delivery (CD). It allows developers to automate the building, testing, and [deployment](#) of [software](#) projects, making the development process more efficient and reliable.

Key features of Jenkins:

- CI/CD: Jenkins supports Continuous Integration and Continuous [Deployment](#), allowing developers to integrate code changes frequently and automate the deployment of applications.
- Plugins: Jenkins has a vast library of plugins that can extend its capabilities. These plugins integrate Jenkins with version control systems (like Git), build tools (like Maven or Gradle), testing frameworks, deployment tools, and much more.
- Pipeline as Code: Jenkins allows the creation of pipelines using Groovy-based DSL scripts or YAML files, enabling version-controlled and repeatable pipelines.
- Cross-platform: Jenkins can run on various platforms such as Windows, Linux, macOS, and others.

Installing Jenkins Locally

Jenkins can be installed on local machines, on a cloud environment, or even in containers. Here's how you can install Jenkins in Window local System environments:

Step-by-Step Guide

1. Prerequisites:

- Ensure that Java (JDK) is installed on your system. Jenkins requires Java 21. If not then [click here](#).
- You can check if Java is installed by running `java -version` in the terminal.

2. Install Jenkins on Window System):

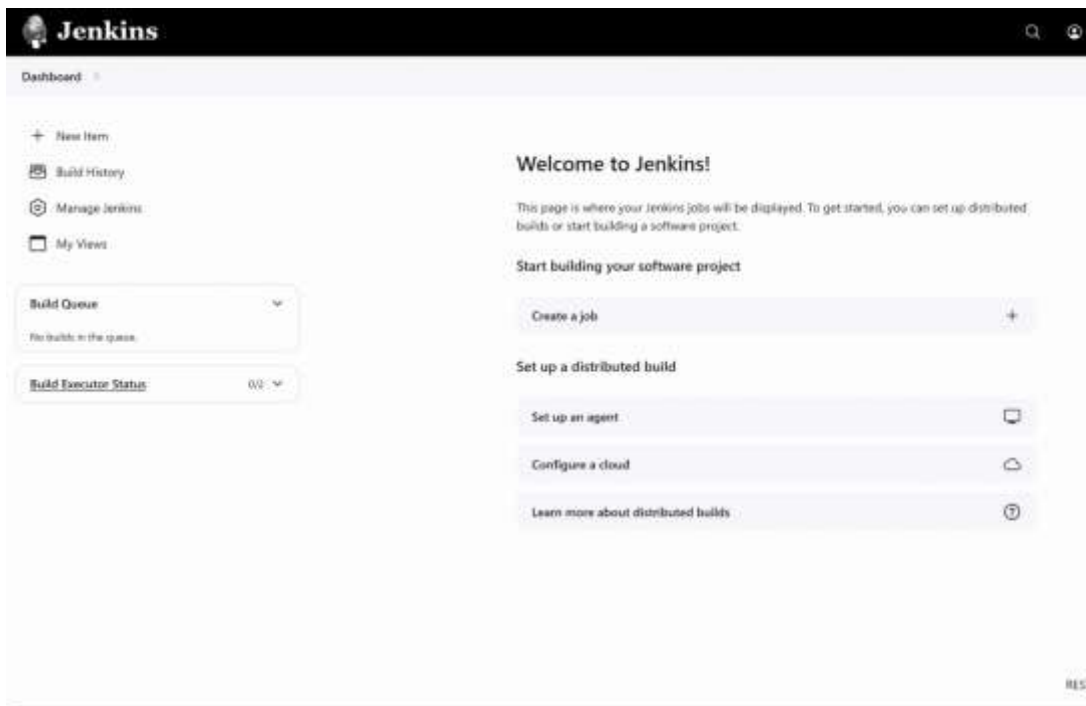
- Download the Jenkins Windows installer from the [official Jenkins website](#).
- Run the installer and follow the on-screen instructions. While installing choose login system: run service as LocalSystem (not recommended).
- After then use default port or you can configure you own port like I'm using port 3030 then click on test and next.
- After then change the directory and choose java jdk-21 path look like C:\Program Files\Java\jdk-21\.
- After then click next, next and then it will ask permission click on yes and it will start installing.
- After successfully installed, Jenkins will be running on port either default port or chosen port like i choose port 3030 by default (you can access it in your browser at `http://localhost:8080`) or `http://localhost:3030`.

Jenkins Setup in browser:

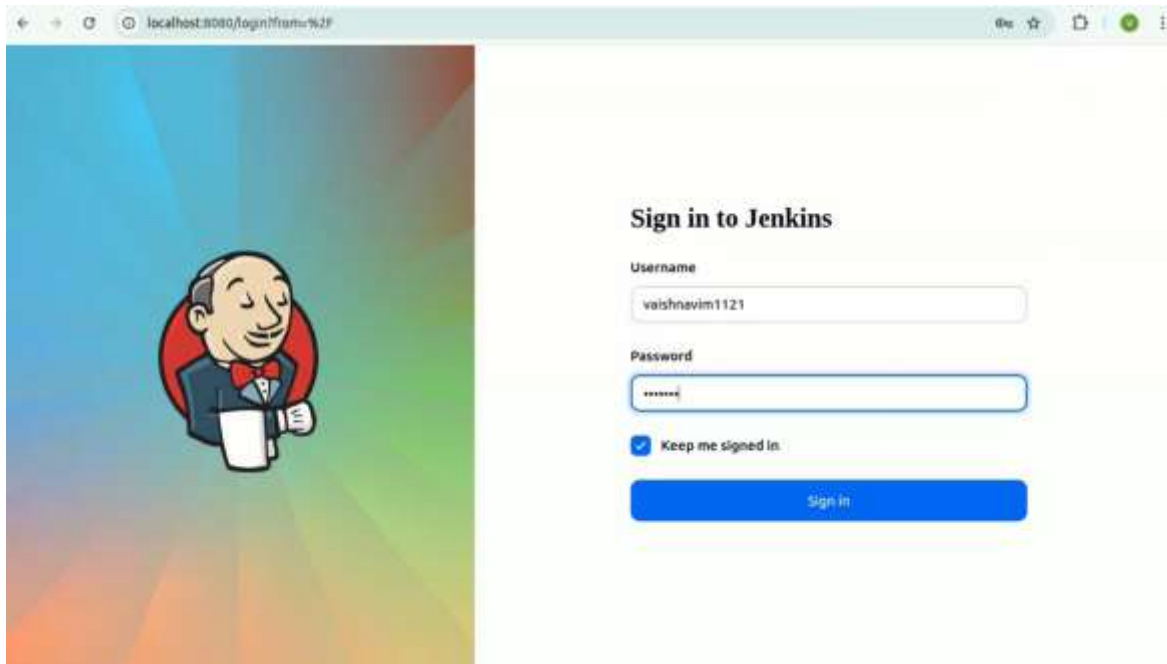
- After opening browser by visiting your local address the browser should look like below screenshot.



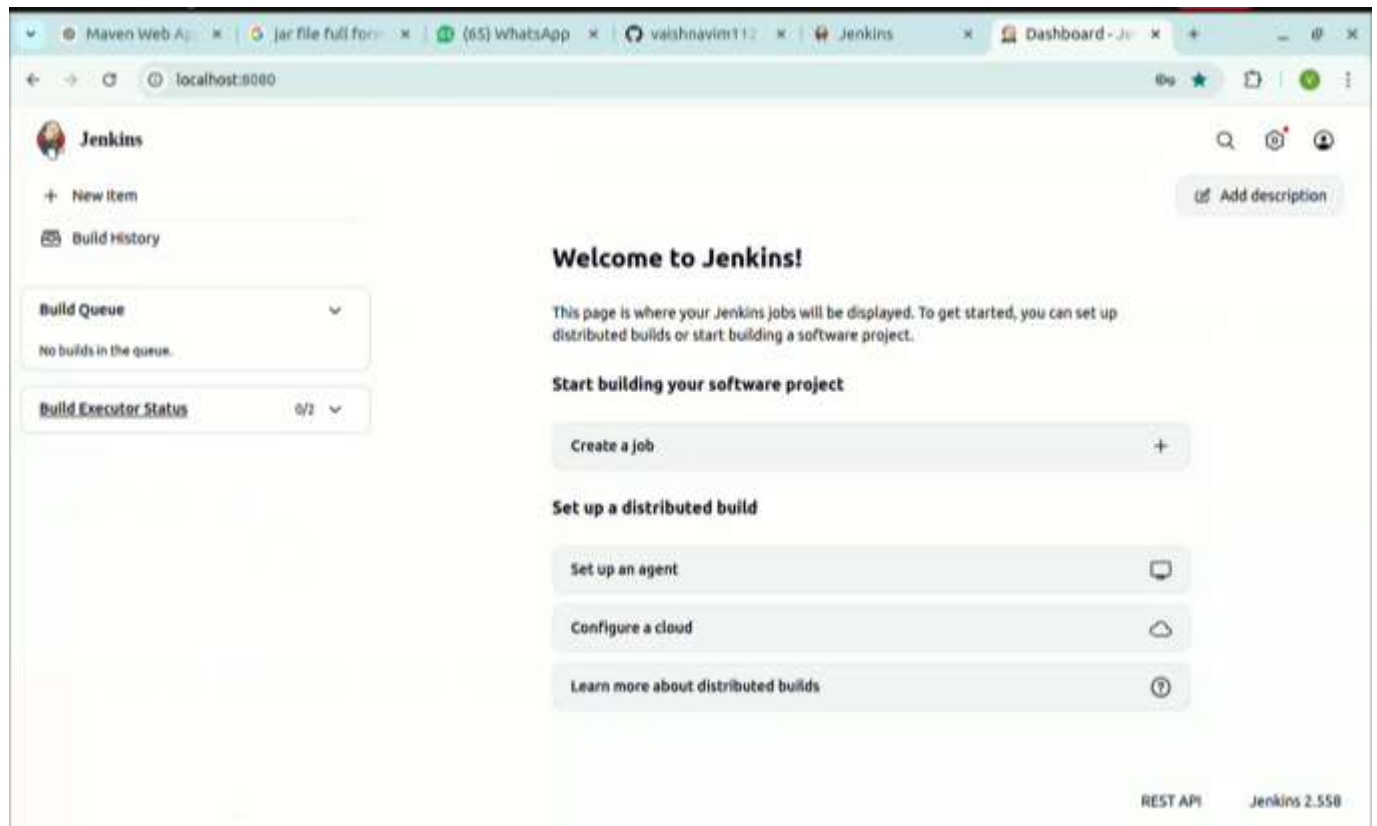
- It will ask administrator password so you have to navigate the above highlighted path and open that initialAdminPassword in notepad or any software to see the password.
- Just copy that password and paste it and click on continue.
- It will ask to customize Jenkins so click on install suggested plugin it will automatically install all required plugin.
- After then create admin profile by filling all details then click on save and continue after then save and finish after then click on start using Jenkin



Enter the login credentials to access the dashboard

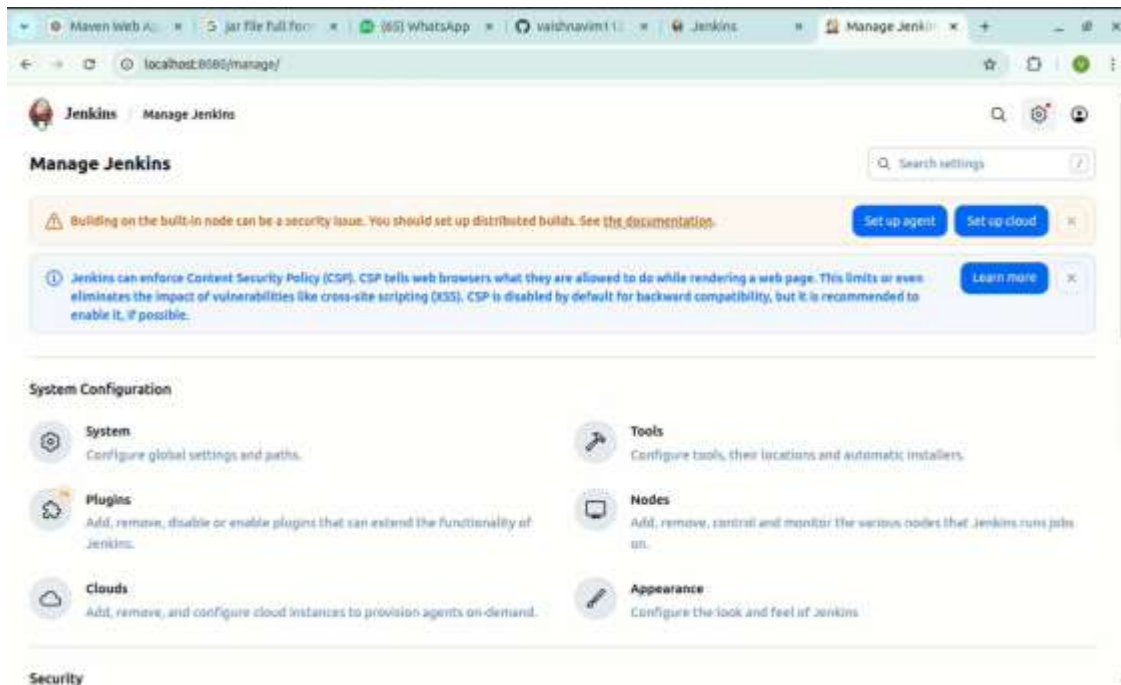


The screenshot shows the Jenkins login page in a web browser. The address bar displays 'localhost:8080/login?from=%2F'. On the left, there is a large blue and red graphic featuring the Jenkins mascot, a man in a suit and bow tie. The main content area is titled 'Sign in to Jenkins'. It contains a 'Username' field with the text 'vaishnavim1121', a 'Password' field with masked characters, a checked checkbox for 'Keep me signed in', and a blue 'Sign in' button.

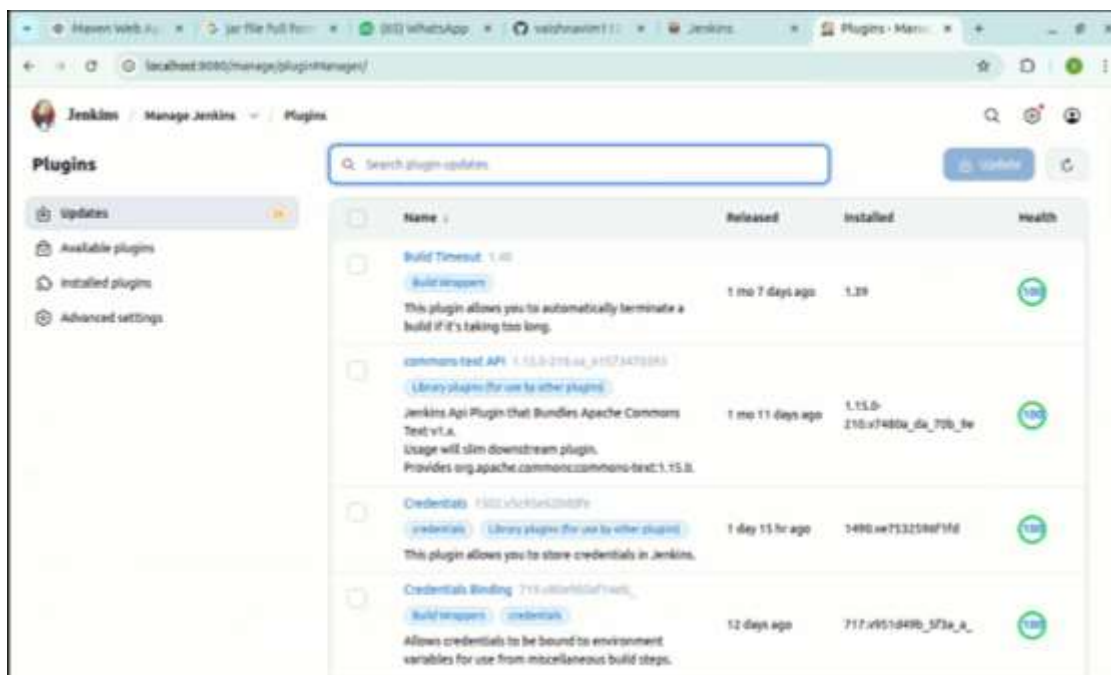


The screenshot shows the Jenkins dashboard in a web browser. The address bar displays 'localhost:8080'. The dashboard has a left sidebar with the Jenkins logo, a 'New item' button, and a 'Build History' link. The main content area is titled 'Welcome to Jenkins!' and includes a welcome message: 'This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.' Below this, there is a section 'Start building your software project' with a 'Create a job' button. Another section 'Set up a distributed build' contains three buttons: 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'. The bottom right corner of the dashboard shows 'REST API' and 'Jenkins 2.558'.

Navigate to Manage Jenkins:



Navigate to Plugins:

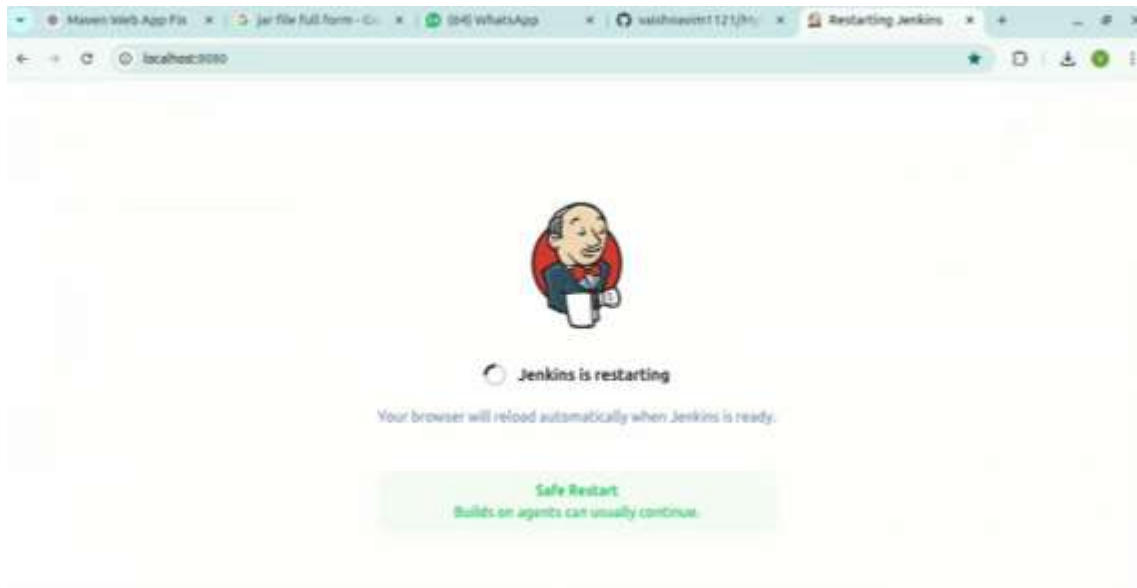
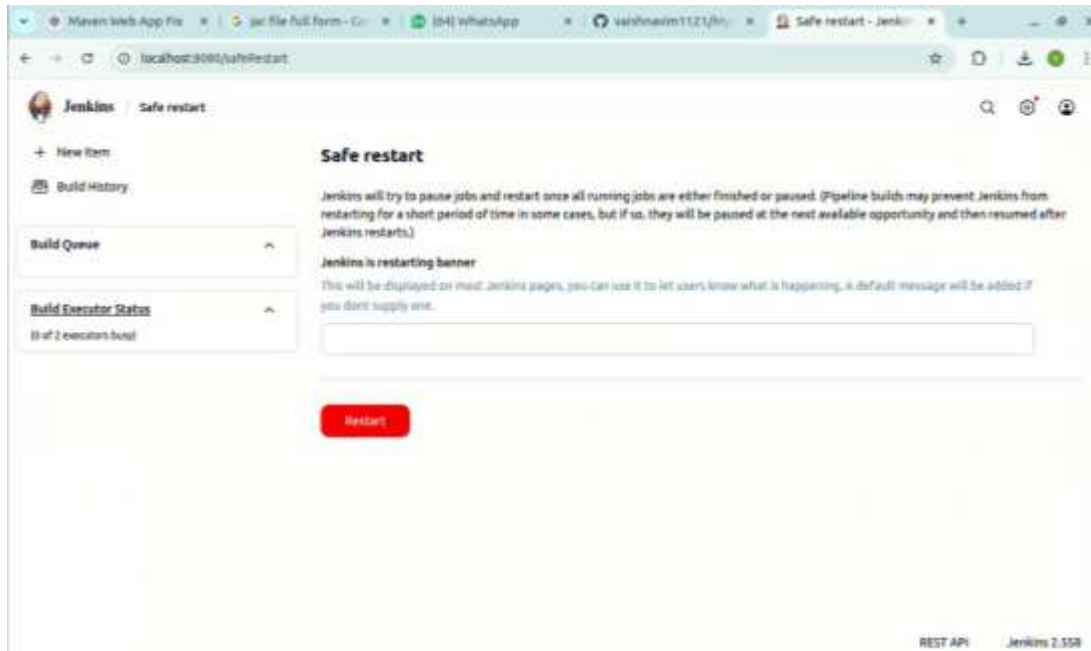


Download the necessary plugins:

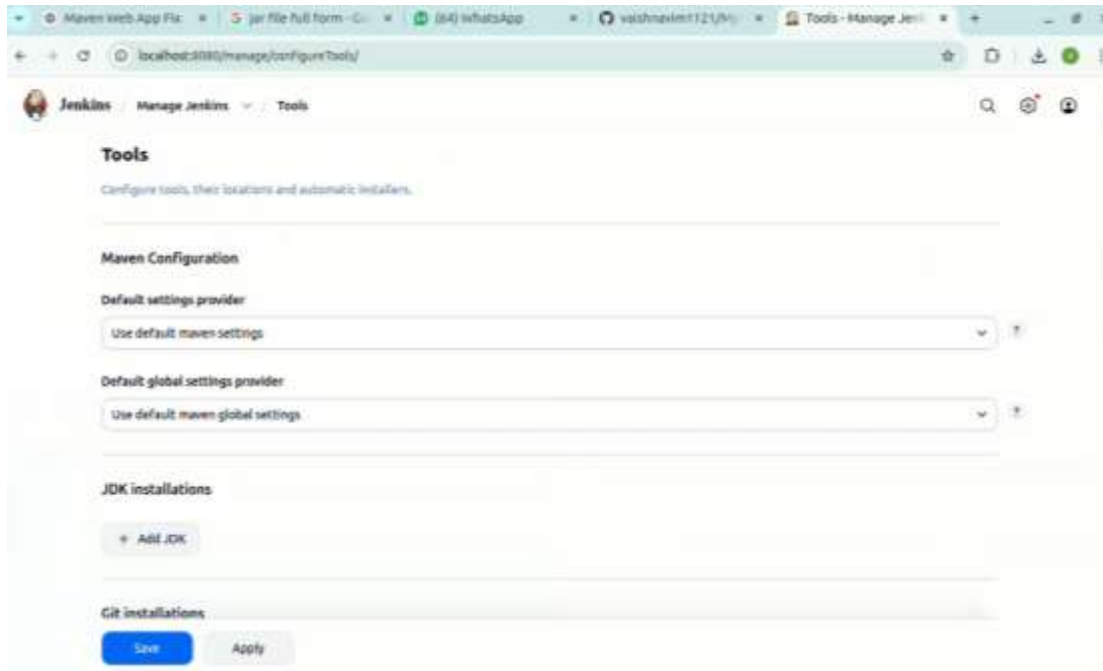
- Maven Integration
- Gradle
- Github Integration
- Pipeline Stage-view Integration

Install the mentioned plugins and verify these downloads in Installed Plugins.

Further, restart the Jenkins using the safeRestart method, i.e, enter the given [url: localhost:8080/safeRestart](http://localhost:8080/safeRestart)



Further, navigate to tools in manage settings



Setup configuration for the downloaded plugins,

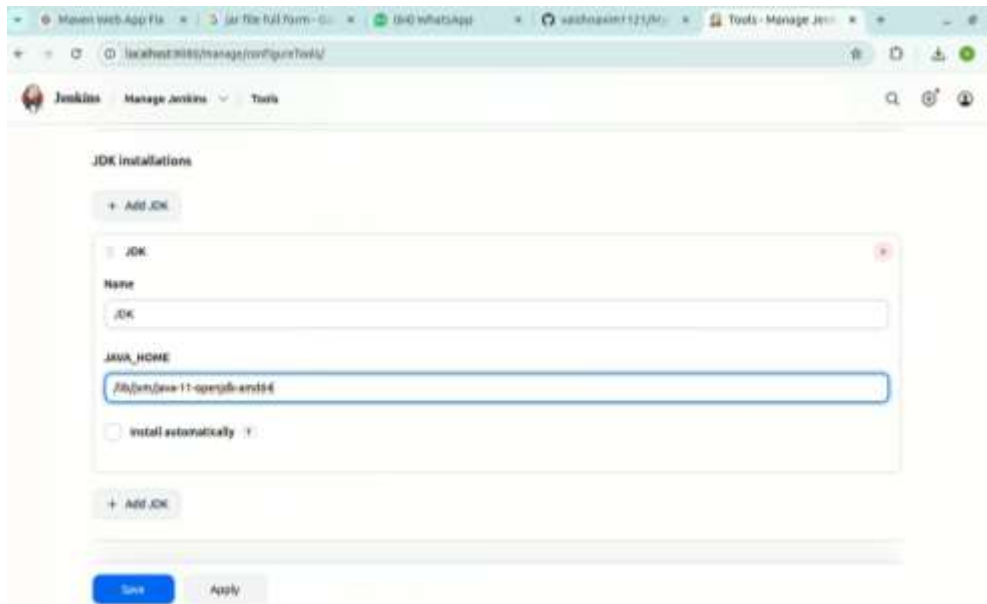
Add JDK

```
vaishnavi@vaishnavi-virtualbox:~/devops21/MyHavenWebApp1$ which java
/usr/bin/java
vaishnavi@vaishnavi-virtualbox:~/devops21/MyHavenWebApp1$
```



```
vaishnavi@vaishnavi-virtualbox: /lib/jvm/java-11-openjdk-am...  
vaishnavi@vaishnavi-virtualbox: /lib/jvm/java-11-openjdk-amd64$
```

Copy the path n paste into Jenkins→Tools



Git configuration

```
/usr/bin/java  
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenWebApp1$ which git  
/usr/bin/git  
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenWebApp1$
```

Git installations

Git

Name

Git

Path to Git executable ?

/usr/bin/git

☐ Install automatically ?

+ Add Git

Gradle Configuration

Gradle

name ?

Gradle

GRADLE_HOME ?

/usr/bin/gradle

☐ Install automatically ?

+ Add Gradle

Maven Configuraton

```
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenWebApp1$ which mvn
/usr/bin/mvn
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenWebApp1$
```

Maven

Name

Maven

MAVEN_HOME

/usr/bin/mvn

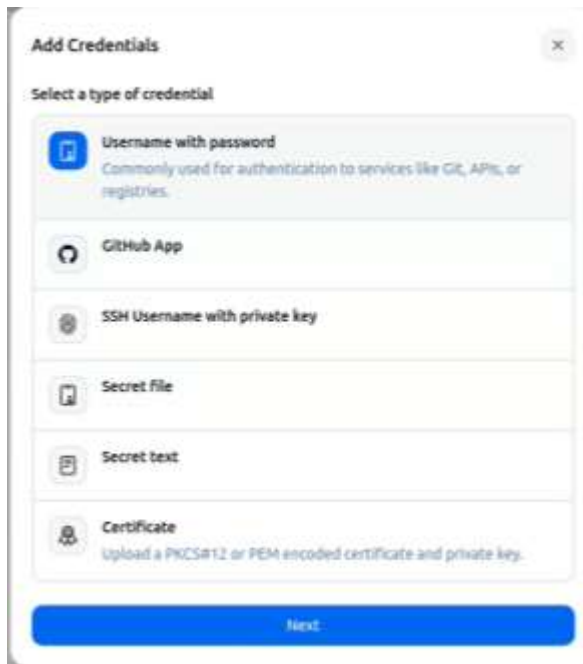
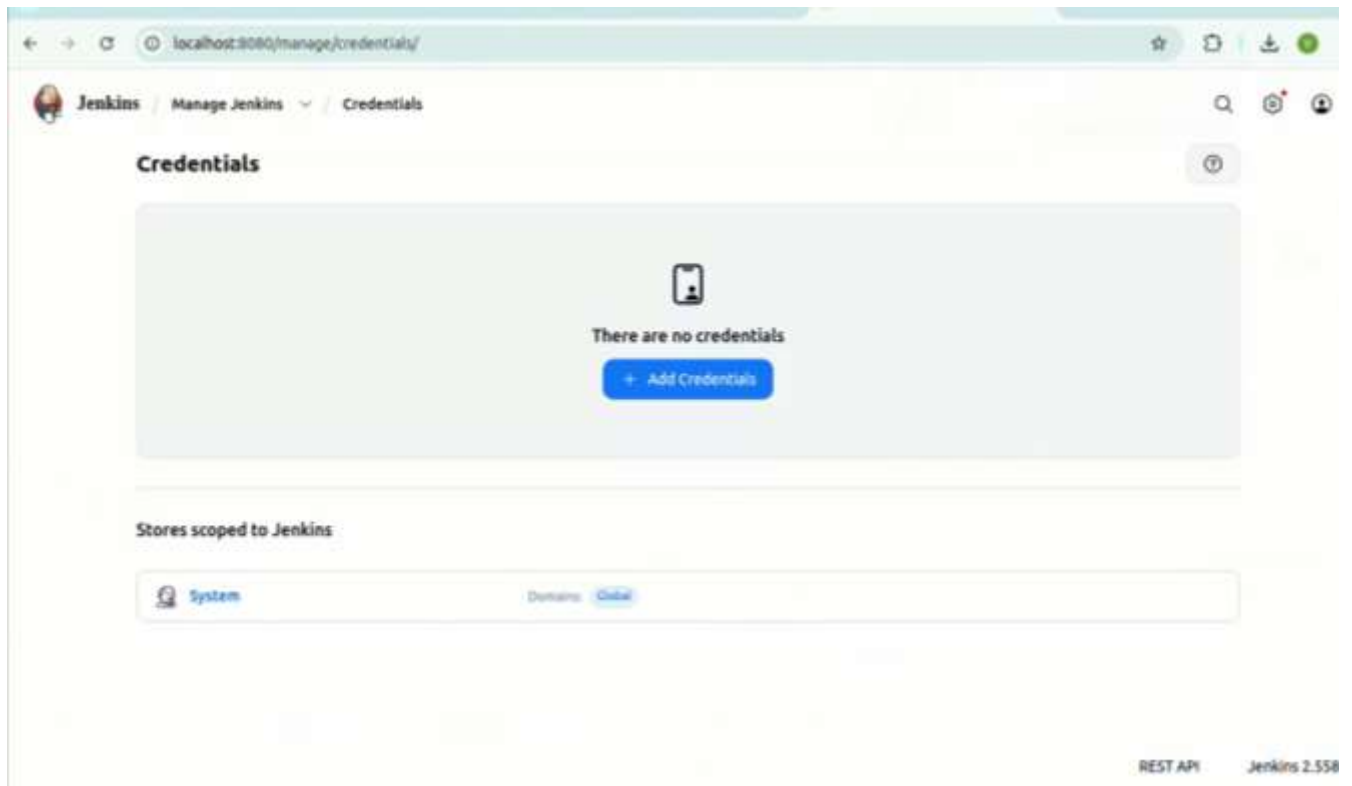
☐ Install automatically ?

+ Add Maven

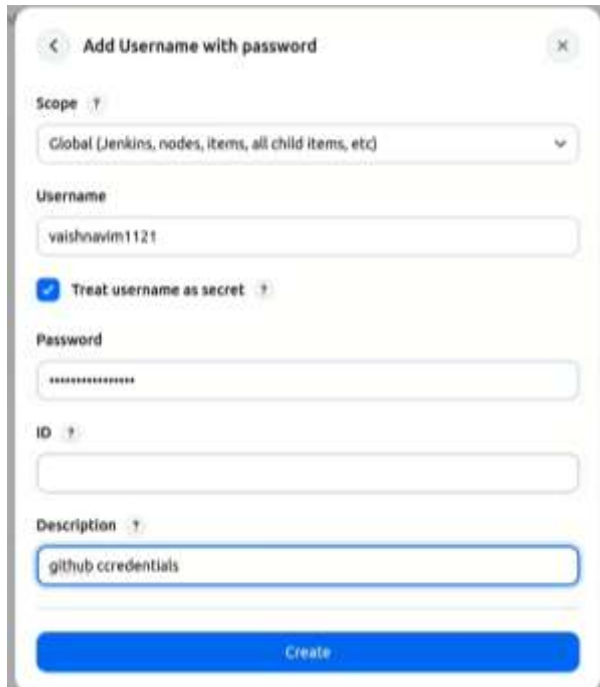
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Add Credentials

Manage Jenkins → Credentials

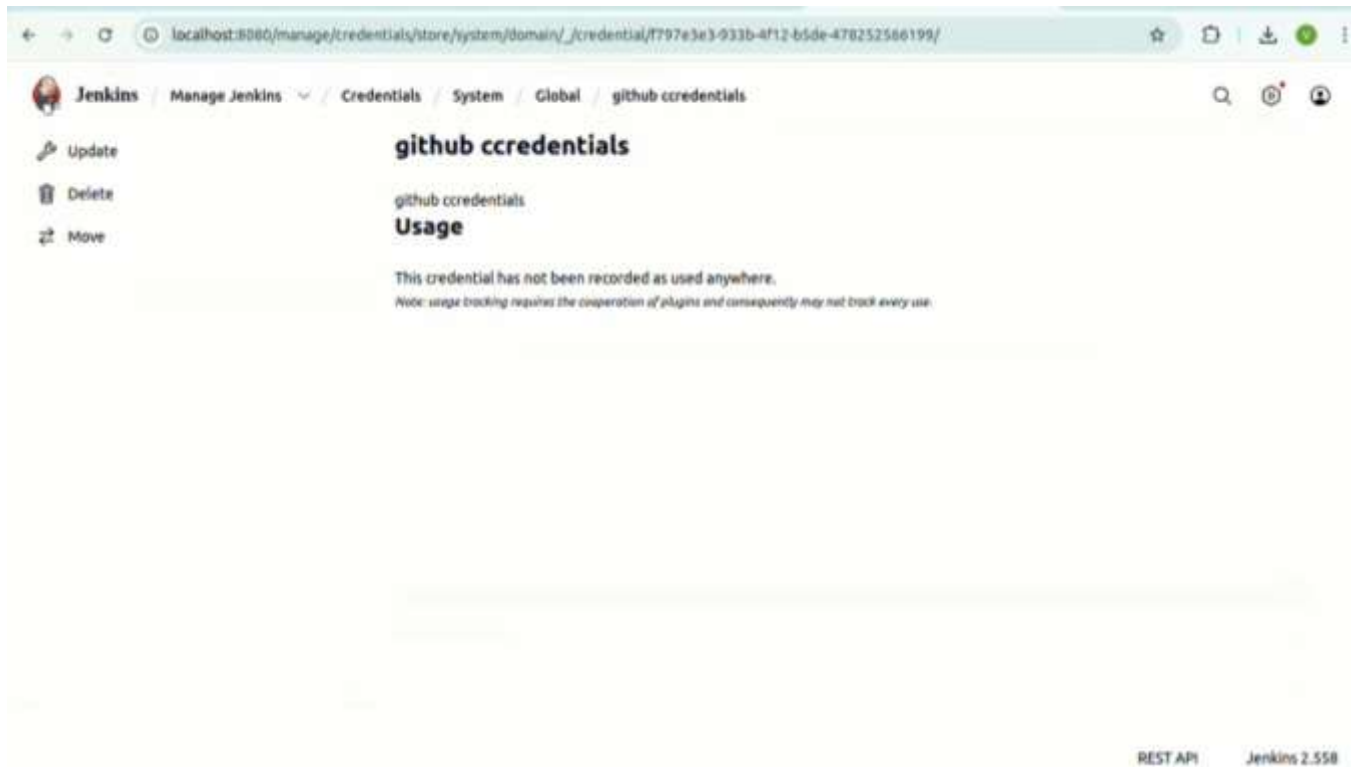


Add username and password of the github credentials



The screenshot shows a modal dialog titled "Add Username with password". It contains the following fields and options:

- Scope:** A dropdown menu set to "Global (Jenkins, nodes, items, all child items, etc)".
- Username:** A text input field containing "vaishnavim1121".
- Treat username as secret:** A checked checkbox.
- Password:** A text input field with masked characters (dots).
- ID:** An empty text input field.
- Description:** A text input field containing "github ccredentials".
- Create:** A blue button at the bottom right.



The screenshot shows the Jenkins web interface at the URL `localhost:8080/manage/credentials/store/system/domain/_/credential/f797e3e3-933b-4f12-b5de-478252566199/`. The breadcrumb navigation is `Jenkins / Manage Jenkins / Credentials / System / Global / github ccredentials`. On the left, there are links for `Update`, `Delete`, and `Move`. The main content area displays the credential name `github ccredentials` and its `Usage`. A message states: "This credential has not been recorded as used anywhere. Note: usage tracking requires the cooperation of plugins and consequently may not track every use." At the bottom right, there are links for `REST API` and `Jenkins 2.558`.

Username

vaishnavim1121

☒ Treat username as secret

Password

Concealed

Change Password

ID

f797e3e3-933b-4f12-b5de-478252566199

Description

github credentials

Save